Fiscal Year:	FY 2019	**Task Last Updated:**	FY 11/14/2019
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Risk of Intervertebral Disc Damage after Prolonged Space Flight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Crew Health NNJ09ZSA002N
Start Date:	07/11/2013	**End Date:**	07/10/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/10/2020 per NSSC information (Ed., 11/12/19) NOTE: End date changed to 7/10/2019 per NSSC information (Ed., 2/11/19)		
Key Personnel Changes/Previous PI:			
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Performance Goal Text:			

Task Description:	Our proposal is a Flight Definition Study that will use state-of-the-art imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crewmembers before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises in space so as to establish pain and disc damage mechanisms that will serve as basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crewmember injury based on clinical findings. We hypothesize that spontaneous space-flight back pain and disc herniation are due to biomechanical and biological pathomechanisms. First, microgravity leads to higher than normal physiologic disc swelling and increased disc height that may stiffen the lumbar motion segment and cause abnormal segmental movement patterns. These biomechanical changes increase risk for annular rupture, vertebral endplate microfracture, and facet joint capsule strain. Second, increased disc swelling may alter nuclear matrix osmotic pressure and nutrient transport from endplate capillaries in adjacent vertebra. These biological changes adversely affect disc cell metabolism, causing pain and inducing disc matrix degradation. Our project directly addresses the Critical Path Roadmap Risks and Questions for NASA regarding disc injury (Integrated Research Plan (IRP) Gap-B4): Is damage to joint structure, intervertebral discs, or ligaments incurred during or following hypogravity exposure? The goal of this research is to characterize space-flight induced changes comprehensively in disc morphology, biochemistry, metabolism, and kinematics. These data will be correlated with measures of back pain intensity and disability. Crewmembers will be imaged pre-flight to establish baseline data and to characterize measurement repeatability. After long-term microgravity exposure (about 180 days on International Space Station-ISS), crewmembers will be studied while maintaining supine posture as soon as possible after return to 1-G in order to quantify the acute effects of prolonged space flight. Also, pre- and post-flight, they will be imaged supine and in standing upright posture at one body weight and again at 1.1 body weight so that MR images are obtained before and after axial body-weight loads. Pre-flight tests will be done within L-120 to L-60 days before the scheduled space flight. Post-flight tests will be undertaken 1-7 days and 30-60 days after landing to investigate re-adaptation to 1-G and to distinguish immediate and longer-term recoveries. Our proposed measures represent a comprehensive set of tests that evaluate exposure severity, potential injury mechanisms, and pain generator localization. Our research will aid understanding of spinal pain and deconditioning during prolonged microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We propose to use state-of-the-art, non-invasive imaging technologies to quantify morphology, biochemistry, metabolism, and kinematics for lumbar discs of crewmembers before and after prolonged space flight. Importantly, we will correlate these data with low back pain that spontaneously arises during prolonged microgravity and after re-adaptation to Earth gravity, so as to establish pathomechanisms that will serve as a basis for future countermeasure development. After successful completion of our investigation, we will deliver a comprehensive database of microgravity-induced intervertebral disc and vertebral changes (type and magnitude) and a prioritization of these changes as to their deleterious effects and risks for crewmember injury based on clinical findings. Importantly, this research will have application to back-pain patients on Earth in general and specifically, to patients exposed to long-term bed rest or lack of mobility (spinal-cord injury patients as well as patients suffering lack of exercise, mobility, and obesity). This research also has application to abnormal spinal curvature and pain suffered by children wearing heavy backpacks to and from school.
Task Progress:	Purpose: Prolonged exposure to microgravity increases risk of disc herniation following space flight. The underlying cause is unknown, but recent work suggests that space flight-induced multifidus atrophy and pre-existing spinal degeneration and pathology may contribute. To understand the relationship between prolonged unloading and elevated risk for post-space flight disc herniation, we longitudinally tracked spinal health of astronauts before and after 6-months in space. Materials and Methods: Twelve NASA astronauts underwent 3T MRI before launch, within a week after return to Earth, and 1-2 months after return to Earth (recovery). MRI scans were used to quantify disc degeneration, disc water content, vertebral endplate irregularities, facet arthropathy, high intensity zones, disc herniation, multifidus cross-sectional area (CSA), and multifidus fat fraction (FF). Results: Five subjects experienced post-flight disc herniation. While subjects had pre-existing spinal degeneration and pathology, there were negligible changes following space flight. For multifidus, both CSA and FF were significantly affected following space flight (-9.4%, p=0.013; +3.2%, p<0.01). Multifidus FF was particularly affected for subjects who experienced disc herniation following space flight (+5.0%, p=0.013). Lastly, pre-existing endplate pathology was associated with insignificant recovery of multifidus FF and was a feature in most subjects who experienced a post-space flight herniation. Conclusion: We found multifidus atrophy is a risk factor for post-flight disc herniation and space flight-induced multifidus atrophy may be exacerbated by pre-existing endplate pathology. This study provides knowledge toward 1) countermeasures to mitigate post-space flight disc herniation and 2) mechanistic relationships between paraspinal muscle health and the development of low back pain that are not observable from cross-sectional clinical studies.
Bibliography Type:	Description: (Last Updated: 08/25/2025)
Articles in Peer-reviewed Journals	Bailey JF, Miller SL, Khieu K, O'Neill CW, Healey RM, Coughlin DG, Sayson JV, Chang DG, Hargens AR, Lotz JC. "From the International Space Station to the clinic: How prolonged unloading may disrupt lumbar spine stability." Spine J. 2018 Jan;18(1):7-14. Epub 2017 Sep 28. https://doi.org/10.1016/j.spinee.2017.08.261 ; PubMed PMID: 28962911; PubMed Central PMCID: PMC6339989 , Jan-2018

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